

Cover Letter — Nature Machine Intelligence (Analysis)

Manuscript: Methodological choices can outweigh algorithmic choices in transcriptomic immunotherapy prediction **Author:** Zhixian Huang (sole author), Putian University — huangzhixian324@gmail.com **Format requested:** Analysis (~2,600 words of main text excluding Methods; 3 figures + 3 tables)

Dear Editors,

I am pleased to submit the above manuscript for consideration in your Analysis format.

Machine learning models that predict treatment response from transcriptomics are proliferating in oncology, and immunotherapy is the fastest-growing application. The evaluation literature, however, keeps producing contradictory rankings: recent cross-cohort evaluations of nine predictors and twenty signatures, a 1,133-patient cross-cancer meta-benchmark, and a 35-task machine-learning benchmark all hold the benchmarking methodology fixed and compare algorithms. None asks the question on which the credibility of this model class depends: **how much of a benchmark's outcome is determined by the benchmarking design itself?**

The manuscript answers that question with a controlled design. Holding six established prediction methods, the patients, and the expression data fixed, it varies only the benchmarking design. Three findings:

1. **Response definition outweighs algorithm choice.** Switching a molecular surrogate endpoint (cytolytic activity) to a clinical endpoint (RECIST v1.1) in the same patients collapses every method whose features overlap the endpoint-defining genes (ElasticNet-Var 0.985 to 0.465; GEP 0.968 to 0.503) — a point estimate $3.5\times$ the largest between-method gap, with the paired-bootstrap caveat disclosed in full (Methods). Two clinical definitions on the same patients move predictions by at most 0.083 (Liu 2019 melanoma) and 0.082 (IMmotion150 renal cell carcinoma) — the collapse is specific to gene-defined surrogates, not to clinical-definition disagreement. The effect replicates in executed external cohorts spanning five cancer types; the protocols were pre-declared and timestamped before the replication data were analyzed.
2. **The circular construction is instantiated practice, not a hypothetical.** A structured field audit (Supplementary Table S30) identifies seven published studies that train or select immunotherapy-response predictors against gene-expression-derived labels in cohorts without ICI outcomes — the immunophenoscore itself was derived against a cytolytic median split. The design rules proposed here therefore bind this literature segment directly.
3. **Statistical floors are quantifiable.** A complete 36-cell permutation-test matrix with multiple-comparison family sensitivity shows that no trainable method is distinguishable from label shuffling on any clinical endpoint at $n \leq 43$; the first detectable signal appears at $n = 73$

(ElasticNet-MI, $p = 0.0232$, $q = 0.028$), seed-stable across 10 regenerated stratified splits, matching an independently anchored power analysis.

The contribution to machine-learning practice is the benchmarking methodology itself: a design-choice decomposition that transfers to any omics prediction benchmark; a reporting convention whose binding rule is the declaration of endpoint-defining genes, with the Circularity Detection Score shipped as an installable package as a null-referenced companion; and a minimum-standards checklist derived from power analysis. All code, data, and per-cell outputs are open, Docker-reproducible, and publicly archived (Zenodo DOI: 10.5281/zenodo.22746254; GitHub, MIT).

Genre precedent. Your 2025 Analysis "Benchmarking AI-powered docking methods from the perspective of virtual screening" (Gu et al., Nat Mach Intell 7:509–520) established that systematic AI-method benchmarks with an evaluation-practice critique belong in this journal's Analysis format; the present manuscript applies the same template to transcriptomic immunotherapy prediction, where the benchmarking-design problem is documented rather than hypothetical.

Two candid notes. The clinical application is immunotherapy rather than basic biology; the methodological contribution, not the clinical setting, is what I believe fits your journal. And the study is single-authored; I know a benchmark asking for community trust has to earn it the hard way, which is why every analysis ran under archived, versioned protocols with a complete re-derivation audit trail.

This manuscript is original, is not under consideration elsewhere, and all authors (one) have approved the submission. I have no competing interests and no funding to declare.

Whether the work belongs here is your judgment alone; I will be grateful for it either way.

Sincerely, Zhixian Huang Putian University huangzhixian324@gmail.com